INCH-POUND

MIL-L-43536F
16 June 1989
SUPERSEDING
MIL-L-43536E
7 October 1982

MILITARY SPECIFICATION

LINER, COLD WEATHER COAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers an insulating liner for the field coat made of quilted polyester batting material.

1.2 Classification. The liner shall be of one type in the following sizes, as specified (see 6.2).

Schedule of sizes

X-Short-X-Small	X-Small
X-Short-Small	Small
X-Short-Medium	Medium
	Large
	X-Large

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Untreated
V-B-871	-	Button, Sewing Hole, and Button, Staple, (Plastic)
V-T-280	-	Thread, Gimp, Cotton, Buttonhole
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-C-508	-	Cloth, Oxford, Nylon, 3 Ounce
MIL-B-41826	-	Batting, Synthetic Fibers: Polyester, (Unquilted and Quilted)
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistance Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-T-43548	-	Thread, Polyester Core: Cotton-, Rayon-, or Polyester-Covered
MIL-T-43566	-	Tape, Textile, Cotton or Polyester, General Purpose, Natural or in Colors

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

THE COLOR ASSOCIATION OF THE UNITED STATES, INC.

Standard Color Card of America

(Standard shades may be available from the Color Association of the United States, Inc., 343 Lexington Avenue, New York, NY 10016-0927. If standard shades are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide samples. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this document may appear in the sample, in which case this document shall govern.

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document.

3.3.1 Basic material. The basic material shall be a quilted polyester batting, conforming to type IV, class 1, style a of MIL-B-41826.

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3.3.2 Facing and binding. The material for facings, neck buttonhole reinforcements and binding shall be nylon oxford cloth, 3.0 ounce, Olive Green 106, conforming to type I, class 2 of MIL-C-508.

3.3.3 Tape. The tape for buttonhole tabs shall be type I, class 3, 1-inch width, dyed Olive Drab 7 conforming to MIL-T-43566.

3.3.4 Labels. Each liner shall have an identification label, a size label, and an instruction label conforming to DDD-L-20. Label color shall be Medium Green, Cable No. 70034, 70130, or 70131 from the Standard Color Card of America. The labels shall show fastness to laundering.

3.3.4.1 Identification label. The identification label shall conform to type VI, class 1.

3.3.4.2 Size label. The size label shall conform to type VI, class 2 and contain the following information for each respective size:

X-Short-X-Small

Height: Up to 63 inches
Chest men's: Up to 33 inches
Chest women's: Up to 36 inches
Stock No.
NATO Size: 5060/7484

X-Short-Small

Height: Up to 63 inches
Chest men's: 33 to 37 inches
Chest women's: 36 to 40 inches
Stock No.
NATO Size: 5060/8494

X-Short-Medium

Height: Up to 63 inches
Chest men's: 37 to 41 inches
Chest women's: 40 to 44 inches
Stock No.
NATO Size: 5060/9404

X-Small

Height: Over 63 inches
Chest men's: Up to 33 inches
Chest women's: Up to 36 inches
Stock No.
NATO Size: 7080/7484

Small

Height: Over 63 inches
Chest men's: 33 to 37 inches
Chest women's: 36 to 40 inches
Stock No.
NATO Size: 7080/8494

Medium

Height: Over 63 inches
Chest men's: 37 to 41 inches
Chest women's: 40 to 44 inches
Stock No.
NATO Size: 7080/9404

Large

Height: Over 63 inches
Chest men's: 41 to 45 inches
Stock No.
NATO Size: 7080/0414

X-Large

Height: Over 63 inches
Chest men's: Over 45 inches
Stock No.
NATO Size: 7080/1424

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3.3.4.3 Instruction label. The instruction label shall conform to type VI, class 3 and contain the following information:

LINER, COLD WEATHER COAT

1. For extra warmth, button into Coat, Cold Weather, Field, or wear as a separate garment in the Parka, Cold Weather, Camouflage.
2. Dry liner and coat separately when damp or wet.
3. Avoid contact with heated stoves, open fires, or other heat sources; fabric may fuse.
4. Wash by hand or machine in warm water with mild soap or detergent, rinse thoroughly, and drip dry or tumble dry, medium setting. DO NOT DRY CLEAN.

DO NOT REMOVE THIS LABEL

3.3.4.4 Combination size and identification label. The identification label (class 1) and the size label (class 2) may be combined into one label conforming to type VI, classes 1 and 2 combined.

3.3.4.5 Combination size, identification, and instruction label. The identification label (class 1), size label (class 2), and instruction label (class 3) may be combined into one label conforming to class 14.

* 3.3.5 Thread. The thread shall be cotton-covered polyester thread, dyed Olive Drab S-1, C.A. 66022, ticket numbers 30, 2 or 3 ply; 50, 1 or 3 ply; and 70, 2 or 3 ply conforming to MIL-T-43548.

3.3.5.1 Colorfastness. The dyed thread shall show fastness to laundering equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to laundering.

3.3.6 Gimp. The cotton gimp for reinforcing buttonholes shall be size No. 8, dyed Olive Drab S-1, Cable No. 66022, conforming to type I or II of V-T-280.

3.3.6.1 Colorfastness. The dyed gimp shall show fastness to laundering equal to or better than the standard sample. When no standard sample is available, the dyed gimp shall show "good" fastness to laundering.

* 3.3.7 Buttons. The buttons shall be 30 line, 4 hole, conforming to type II, class D, style 26 of V-B-871. The color shall be Olive Green BP and the finish shall be dull.

3.4 Figures. Figure 1 is furnished for information only. Should variation from the written specification appear in figure 1, the written specification shall govern.

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3.5 Patterns. Standard patterns, which provide a 3/8 inch seam allowance for all seams, except the facing and neck buttonhole reinforcement seams which shall be 1/4 inch, will be furnished by the Government. The standard patterns shall not be altered in any way and shall be used as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the standard patterns.

3.5.1 Pattern parts. The pattern parts and applicable material for cutting the liner parts shall be as specified in table I.

TABLE I. List of pattern parts

Materials	Nomenclature	Cut parts
Quilted polyester batting	Body	1
	Top sleeve	2
	Under sleeve	2
Cloth, nylon, oxford	Neck buttonhole reinforcement	1

3.6 Construction.

3.6.1 Stitches, seams, and stitching. Stitch, seam, and stitching types specified in table II shall conform to FED-STD-751. Where two or more seam or stitching types are given for the same operation, any one of them may be used. Seam allowances shall be maintained with seams sewn so that raw edges, run-offs, twists, pleats, puckers, or open seams will not result. Thread tension shall be maintained so that there is no tight or loose stitching. All seams shall start and finish evenly.

3.6.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/4 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.6.1.1.1 Repairs of type 301 stitching.

3.6.1.1.1.1 Quilted batting stitching. Broken stitches of the quilting pattern may be repaired by stitching back of each end of break not less than 1 inch.

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3.6.1.1.1.2 Liner fabrication stitching.

a. When thread breaks or bobbin run-outs occur during stitching, except prestitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Except for prestitching, thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. When damaged thread has been removed, needle holes do not constitute damage to the material. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.6.1.2 Type 301, 401, 504, 505, 515, 516, or 519 stitching. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn. All repairs shall be in accordance with 3.6.1.1.1.2.a. and 3.6.1.1.1.2.b. Type 301 stitching may be used to repair type 401 stitching.

3.6.2 Buttonhole. The buttonholes shall be eyelet-end, taper-bar type, worked over gimp (see 3.3.6) with not less than four tacking stitches at bar end (not counting crossover stitches). The purling shall be on the facing side and reinforcement side. The finished cut lengths shall be 3/4 to 7/8 inches. The buttonholes shall be clean cut with the stitching securely caught in the fabric. Additive treatments or materials used to secure the ends of the stitching or to stiffen the fabric, prior to stitching the buttonholes are prohibited.

3.6.3 Bartacking. Unless otherwise specified, bartacking shall be as follows:

Length	Width	Stitches per bartack
1/2 inch ± 1/16 inch	1/8 inch ± 1/32 inch	28
3/4 inch ± 1/16 inch	1/8 inch ± 1/32 inch	42

Bartacking shall be free from thread breaks and loose stitching.

3.6.4 Automatic stitching. Automatic stitching machines may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three or more tying, overlapping or back stitches are used to secure the ends of the stitching.

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3.6.5 Thread ends. All thread ends shall be trimmed to a length of not more than 1/4 inch.

3.7 Repairs. Repairs such as mends, darns, patches, or splices are not permitted on the liner.

3.8 Manufacturing operations requirements The liner shall be manufactured in accordance with the operations requirements specified in table II. The contractor is not required to follow the exact sequence of operations listed and additional basting or holding stitching to facilitate manufacture is permissible provided the thread is removed or does not show on the finished liner.

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TABLE II.

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut the liner parts in strict accordance with patterns furnished which show size, directional lines (warp or filling), and notches for proper assembly of parts. NOTE: Care shall be exercised when cutting the basic material and handling the cut parts because the ends of the quilting stitching have a tendency to run back. b. Cut the oxford material for the neck buttonhole reinforcement in accordance with pattern furnished which shows size and directional lines. c. Cut the nylon oxford for the facing in accordance with the requirements specified in operation 6. d. Cut the binding on the straight or bias in sufficient width to meet the requirements specified in operation 12. Binding may be placed on rolls. e. Cut the tape for sleeve buttonhole tabs to meet the finished length specified in operation 10.						
2.	Replacement of defective components. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2 shall be removed from production and replaced with nondefective and properly matched components.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Replacement of defective components. (cont'd) NOTE: Broken stitches of the quilting pattern may be repaired by stitching back of each end of break, not less than 1 inch.						
3.	Component marking. Mark and ticket all components of the liner to ensure a uniform size throughout the liner. Any method of marking may be used except: a. Metal fastening devices. b. Sew on type tickets. c. Adhesive type tickets which leave traces of adhesive on the material upon removal of the ticket.						
4.	Labels. Sew instruction, size, and identification labels 1/8 inch ± 1/16 inch from edges, on all four sides to inside center (± 1/2 inch) of back part as follows: a. Size label. May abut or be caught in the line of stitching of the neck buttonhole reinforcement piece. b. Identification label. May abut or be caught in the line of stitching of the size label.	301		10-14	50	50	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Labels. (cont'd)						
	c. Instruction label. May abut or be caught in the line of stitching of the identification label or combination label (when applicable).						
	d. Combination label (when applicable). Shall be positioned as previously specified in (a).						
5.	Attach neck buttonhole reinforcement.						
	Finished appearance. Each liner shall have the two neck buttonholes reinforced on inside of liner with the nylon oxford material.						
	a. Turn under side edges and bottom edges of the reinforcement 1/4 inch. Position reinforcement as indicated by marks on pattern and stitch through liner, 1/16 to 1/8 inch from turned edges.	301 or 401	LSd-1	10-14	50 50	50 50	
6.	Seam facings to right and left fronts.						
	a. The facing shall be 1-3/4 inches ± 1/8 inch wide and of a sufficient length so as to cover the full length of the front edge of the front piece for the width of the facing. The length of the facing shall be cut in the warp direction of the cloth. Position the facing on the inside of the front piece with the edge of the facing even with the front edge of the front piece. Turn in the back edge of facings and single stitch 1/16 to 1/8 inch from turned edge.	301 or 401	LSd-1	10-14	50 50	50 70	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Seam facings to right and left fronts. (cont'd)						
	b. Stitch front edge of facings to front edge of front parts.	301 or 401	SSa-1	10-14	50 50	50 70	
	or						
	c. Front edge of facings may be caught in overedge stitching of front parts (see operation 7).						
	d. Trim the top and bottom edges of the facing to match the underlying edges of the front pieces.						
	or						
	e. Turn in back raw edges of facing 1/4 inch, position on inside of front parts respectively, with front edges of facings even with front edges of fronts. Double stitch facings to front with a 1-1/8 to 1-3/8 inch gage.	301 or 401	LSd-1 and SSa-1	10-14	50 50	50 70	
7.	Overedge stitch liner parts.						
	Overedge stitch all raw edges of liner except edges at shoulders and edges joining top and under sleeves.	504 or 505	EFd-1	6-12	70 70	70 70	
8.	Bind armholes.						
	Bind armhole openings with the binding starting and finishing at edges of shoulder seam allowances.	301	BSc-1	10-14	50	50	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Join shoulder seams and sleeve seams.						
	Finished appearance. The overedge stitched edges shall finish on the outside of the liners.						
	a. Join the top sleeve to undersleeve and join shoulder seams by overedging and seaming the raw edges together in one operation. The gage of overedge stitching shall be 3/16 to 1/4 inch.	515 or 516 or 519	LSbm-3(a)	10-12	50 70	50 70	chain stitching all overedge stitching threads
	b. Turn seam allowance to one side and top stitch 3/16 to 1/4 inch from folded edge.	301 or 401	LSbm-3(b)	10-14	50 50	50 50	
10.	Make sleeve buttonhole tabs.						
	Finished appearance. Each liner shall have four buttonhole tabs, measuring 2-7/8 inches ± 1/4 inch in length when finished and attached to sleeve.						
	a. Turn in one raw end of each tab 1/4 to 3/8 inch and stitch or bartack 3/16 to 1/4 inch from folded edge across width of tape.	301 or 3/4 inch bartack	EFa-1	10-14 42 per bartack	50 50	50 50	
	b. Make a vertical buttonhole in the center (± 1/8 inch) of each tab, with the inside edge of eyelet 1/2 to 5/8 inch from the turned end of tab.	Buttonhole		52-56 per buttonhole including tack	50	50	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
*11.	Attach sleeve buttonhole tabs.						
	Center a buttonhole tab on each backarm and underarm seam of both sleeves on the inside (1/4 inch off center permitted) with raw edges aligned and the finished end of tab toward top of sleeve, and single stitch to bottom of sleeve, across the width of tab, 1/8 to 1/4 inch from edge. Purling of buttonholes shall face the inside of the sleeve when tab is extended below bottom edge of sleeve.	301 or 401	SSa-1	10-14	50 50	50 70	
12.	Bind edges of liner.						
	Finished appearance. The binding shall finish 3/8 to 5/8 inch wide. The raw edge of the neck buttonhole reinforcement and raw ends of sleeve buttonhole tabs shall be caught in the binding stitches.						
	a. Bind overedge stitched edges of liner all around outside edges, bottom of sleeves, and top of sleeves.	301	BSc-1	10-14	50	50	
	b. The raw finished ends of binding shall be turned in and caught in the stitching; or may be turned over binding to the outside of liner and tacked or bartacked.	301 or 1/2 inch bartack	SSa-1	10-14 28 per bartack	50 50	50 50	
*13.	Join sleeves to liner.						
	a. Lap sleeves on armhole opening on outside of liner with edges of binding aligned (± 1/16 inch), and double stitch sleeves to liner, from notch to notch leaving armpit open, with a 5/16 to 3/8 inch gage.	301	LSa-2	10-14	50	50	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
13.	Join sleeves to liner. (cont'd)						
	b. Bartack across width of binding at ends of stitching on front and back part of sleeve joining seams.	1/2 inch bartack		28 per bartack	50	50	
14.	Mark and make buttonholes.						
	Finished appearance. The buttonholes on the fronts of the liner shall be vertical, through facings, and 7/8 to 1 inch from front edge. The two neck buttonholes on back shall be at right angles to the neck edge, positioned as indicated by marks on pattern with the inside edge of eyelet 3/4 to 7/8 inch from top edge of liner.						
	a. Make 8 buttonholes, three on each front and two in the neck at back as indicated by marks on pattern. The eyelet end of the buttonholes along front edges may be up or down, i.e., on one front the eyelet end may be up and on the other front, the eyelet end may be pointed down.	Buttonhole		52-56 per buttonhole including tack	50	50	
15.	Clean liner.						
	a. Trim and remove all thread ends, except ends of the quilting stitching on the basic material.						
	b. Remove all spots and stains.						
	c. Remove all size markings.						
*16.	Attach buttons.						
	Attach three buttons to front of garment per marks on patterns.	101 or 301		14-16	30	30	

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3.8 Measurements. The liner shall conform to the finished measurements in table III.

TABLE III. Finished measurements (inches)

Sizes	1/2 Chest 1/	Back length 2/	Sleeve length 3/
X-Short-X-Small	22-1/4	26	21
X-Short-Small	24-1/4	26-1/2	21-1/4
X-Short-Medium	26-1/4	27	21-1/2
X-Small	22-1/4	27-1/2	22
Small	24-1/4	28	22-1/4
Medium	26-1/4	28-1/2	22-1/2
Large	28-1/4	29	22-3/4
X-Large	30-1/4	29-1/2	23
Tolerance	± 3/4	± 1/2	± 1/2

1/ 1/2 chest measurement taken with the liner flat and the front edges abutting, from folded edge to folded edge at the base of armhole.

2/ Back length measurement taken along center back from edge of neck to bottom edge of liner.

* 3/ Measure overedged side of backarm seam from top of binding at armhole to bottom of binding at wrist with no tension applied to the seam. The measurement is to be taken at the center portion of the seam allowance.

3.9 Workmanship. The finished liner shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item visual examination. The end items shall be examined for the defects listed in table IV. The lot size shall be expressed in units of liners. The sample unit shall be one liner. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects, 15.0 for major and minor A combined defects, and 40.0 for total (major, minor A and minor B combined) defects.

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TABLE IV. End item visual defects

Examine	Defect	Classification Major	Minor A	Minor B
Material defects and damages				
* Quilted batting:	Any hole, cut, tear, needle chew, smash, or multiple float (in any ply)	101		
	Darn or mend (in any ply)		201	
	Exposed drill hole (in any ply)			301
	One or more rows of stitching omitted in the quilting	102		
	More than 1-1/2 inches of broken or missing stitches in the quilting			302
	Thick or thin place	103		
	Piecing or splicing of outer covering (either ply)	104		
	Abrasion in either ply of outer covering resulting in a weak place, or any weak place	105		
Oxford cloth:	Any hole, cut, or tear	106		
	Shade bar, fine or coarse filling bar (oxford cloth)			303
	Abrasion mark			304
Cleanness	Any spot or stain			305
	Thread ends not trimmed as specified, or loose threads not removed on major portion of liner, except thread ends of quilting stitching on basic material			306
	One or more size markings not removed, when applicable			307
	Any size markings exposed or visible on outside of liner			308
Components and assembly	Any component part or required operation omitted (unless otherwise classified herein)	107		
	Any operation not performed as specified (unless otherwise classified herein)			309

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Cutting	Any component part not in accordance with directional lines indicated on pattern or not cut in accordance with specification requirements	108		
Seams and stitching	Accuracy of seaming: -seams twisted, puckered, or pleated			310
	NOTE: Nylon fabric has a tendency to pucker at joining seams and at stitchings. The puckering, when not so excessive as to seriously affect appearance, shall not be classified as a defect.			
	Part of liner caught in any unrelated operation or stitching	109		
	-end of seam or stitching produced with 301 stitch type, when not caught in other seams or stitching, backtacked less than 1/4 inch		202	
	-stitching overlapping end of thread break less than 1/2 inch			311
	-ends of a continuous line of stitching not overlapped or overlapped less than 1/2 inch			312
	Gage of stitching and seam allowances: -irregular i.e., unevenly gaged or corresponding stitchings not uniformly gaged (to be scored only when the condition exists on major portion of seam)		203	
	-width not as specified or not within range of gage specified or varies more than 1/16 inch when no range is specified			313
	-edge or raised stitching sewn too close to edge resulting in damage to cloth	110		

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)	Open seams: (not applicable to quilting stitching)			
	Any seam open by:			
	-up to 1/4 inch inclusive			314
	-more than 1/4 inch to 1 inch inclusive		204	
	-more than 1 inch	111		
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off stitches occur. On double stitched seams, a seam is classified as open when either one side or both sides of seam are open.			
	Raw edge (excluding overedge stitching and end of binding, when not turned under):			
	-1/8 up to 1/4 inch inclusive			315
	-more than 1/4 and up to 1 inch inclusive		205	
	-more than 1 inch	112		
	Any raw edge caused by omission of overedge stitching for a distance of 3 inches or more		206	
	NOTE: Raw edges are classified as such when they occur at places where edge is required to be bound, turned under, or overedge stitched.			
	Run-off: Joining seam - When resulting in an open seam use "Open seam" classification.			
	Edge or raised stitched (when not resulting in an open seam):			
	-1/4 to 1/2 inch inclusive			316
	-more than 1/2 inch		207	

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)	Seam and stitch type:			
	-seam type not specified		208	
	-stitch type not as specified		209	
	-overedge stitched edge on joining seams not finishing on outside of liner		210	
	Stitch tension:			
	-loose tension resulting in a loose seam	113		
	-loose tension on raised or edge stitching		211	
	-tight tension (stitches break when strain is applied to seam or stitching	114		
	NOTE: Puckering is evidence of tight tension. When puckering is evident, the seam shall be tested by exerting pull in lengthwise direction of seam.			
	Stitches per inch (to be scored only when the condition exists on the major portion of seam except labels):			
	Less than minimum specified:			
	-one or two stitches		212	
	-three or more stitches	115		
	One or more stitches less than minimum specified on one or more labels		213	
	One or more than the maximum specified			317
Bartack	Not specified size or type			318
	Loose stitch tension on one or more			319
	Missing, insecure or misplaced, not serving intended purpose, unless otherwise specified herein:			
	-one bartack			320
	-two or more bartacks		214	

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Buttonholes (construction)	Stitching not caught on fabric, causing stitching to pull away from fabric	116		
	Wrong type, incomplete stitching, or uncut buttonhole		215	
	One or more broken stitches in:			
	-one buttonhole		216	
	-two or more buttonholes	117		
	More than two skipped stitches in:			
	-one buttonhole			321
	-two or more buttonholes		217	
	Gimp omitted in one or more buttonholes			322
	Buttonhole less than 3/4 or more than 7/8 inches long		218	
	Purling finishing on outside of liner			323
	End stitched with less than 4 stitches per bar end (see 3.6.2):			
	-one buttonhole		219	
	-two or more buttonholes	118		
	One or more buttonhole(s) omitted	119		
Binding	Not caught on fabric, i.e., can easily be pulled away from edge of fabric	120		
	End of binding not turned over the binding to outside of liner and tacked, or not turned under and caught in the stitching			324
	Binding stretched, i.e., edges having a wavy appearance			325
	Finishing less than 3/8 or more than 5/8 inch wide			326

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Front facing	Stitched on outside of liner	121		
	Front, top, or bottom edge not completely caught in binding stitching		220	
	Back edge not turned under and stitched		221	
	Tight, short, or twisted, causing excessive fullness or twist on front			327
* Sleeve	Liner lapped on sleeves		222	
	Front edge of binding on sleeve not even with back edge of binding ± 1/16 inch of armhole of liner			328
	Bartack at end of sleeve setting seam: -misplaced not serving intended purpose or missing	122		
	-not extending full width of binding			329
* Buttonhole tab	One or more omitted, or not caught in binding stitching	123		
	Less than 2-5/8 or more than 3-1/8 inches long			330
	Placed more than 1/4 inch from center of back seam or underarm seam of sleeves			331
	Buttonhole tab placed at an excessive angle causing tab to lie in a non-vertical position. Free end of tab more than 1/2 inch from the center of sleeve seam			332
	Finished end of tab not toward top of sleeve			333
	One or more tabs not positioned on the inside of the liner		223	
	Bottom hem of tab less than 1/4 or more than 3/8 inch in length			334

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Front buttonhole (position)	Less than 7/8 or more than 1 inch from front edge		224	
	Constructed in horizontal position			335
	Vertical position off center by 1/4 inch or more		225	
Neck buttonhole (position)	One or more not through reinforcement piece		226	
	Back buttonhole less than 3/4 or more than 7/8 inch from top edge of liner			336
	Eyelet end not toward neck edge of liner on back			337
Sleeve tab buttonhole (position)	Less than 1/2 or more than 5/8 inch from finished bottom end		227	
	Off center by more than 1/8 inch			338
	Purling of buttonhole not toward the inside of liner when tab is extended below bottom edge of sleeve			339
* Buttons	Missing, loose, or broken:			
	-two or more	124		
	-one		228	
	Button, misplaced by more than 1/4 inch		229	
	Stitching not locked at end of cycle (pull at loose end of thread when accessible to see if it will ravel)		230	
Labels	Size, instruction, identification, or combination label:			
	-missing, incorrect, or illegible	125		
	-not positioned as specified		231	
	-not stitched on four sides			340
	-off center on back by more than 1/2 inch			341

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4.4.3 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified in table III. Any dimension deviating from the specified requirement and sleeves uneven in length by 3/4 inch or more shall be classified as defects. The lot size shall be expressed in units of liners. The sample unit shall be one liner. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

* 4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required Number of liners per bundle is more or less than required 1/ Size shown on one or more liners not as specified on container 1/

1/ For this defect, two bundles from each container in the sample shall be examined.

* 4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified

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Examine	Defect
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Each liner shall be laid flat, front up with front edges abutting. The sleeves shall be folded at right angles across the chest. For the larger sizes, it may be necessary to adjust the sides in order to attain the desired width of approximately 23 inches. The liner shall then be folded in half the length. The completely folded liner shall measure approximately 23 by 14-1/2 inches. Five folded liners of one size only shall be neatly stacked in a bundle and securely tied with cotton tape or twine approximately 4 inches in from each end.

* 5.1.2 Commercial preservation. The liners shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

* 5.2.1 Level A packing. Fifteen liners of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A bundles shall be packed flat three in depth within a shipping container. Inside dimensions of each shipping container shall be approximately 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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* 5.2.2 Level B packing. Fifteen liners of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A bundles shall be packed flat three in depth within a shipping container. Inside dimensions of each shipping container shall be approximately 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

* 5.2.3 Commercial packing. Liners, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), liners, packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

* 5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The liner is intended for wear by military personnel of the Department of Defense as a component of the cold-dry uniform, in conjunction with the Coat, Cold Weather Field, and also can be worn in conjunction with the Parka, Cold Weather, Camouflage. The liner may be buttoned into a field coat or worn as a separate buttoned front liner.

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* 6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

* 6.5 Subject term (key word) listing.

Apparel
Clothing
Outerwear

6.6 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:

Army - GL
Navy - NU
Air Force - 11

Review activities:

Army - MD
Navy - MC
Air Force - 82, 99
DLA - CT

User activity:

Air Force - 45

Preparing activity:

Army - GL

(Project 8415-0675)

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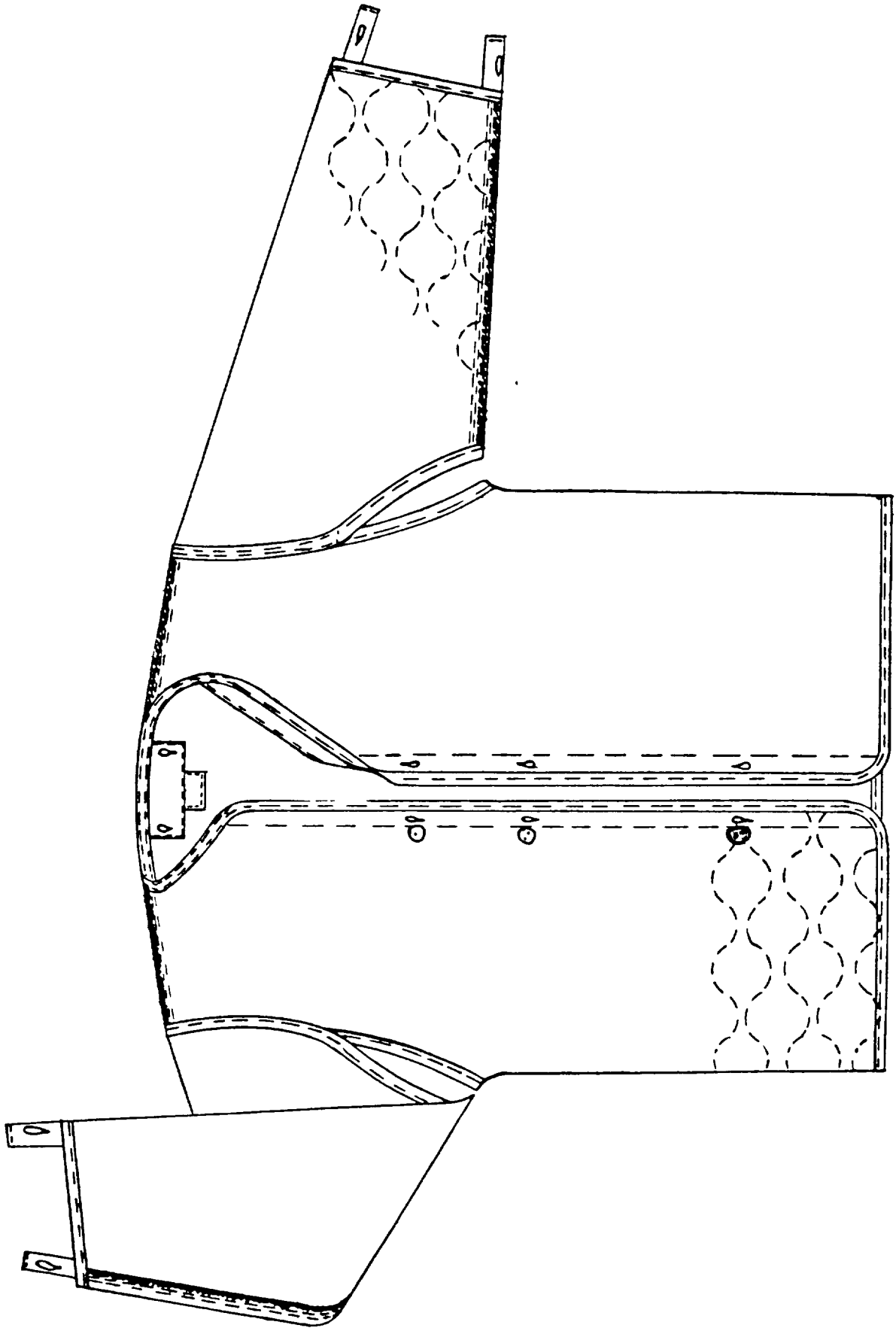


FIGURE 1. Liner, Cold weather coat.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-L-43536F	Liner, Cold Weather Coat

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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